

Slide 1: Title Slide

Good morning, everyone. My name is Lin Zhi-Long. I am a Master's student in the AI department at Taiwan Tech. Today, I will talk about "State-of-Health Estimation for Lithium-ion Battery Packs." This presentation comes from a recent paper in the Journal of Energy Storage.

Slide 2: Outline

Here is my plan for today. First, I will explain why this topic is important. Second, we will look at how batteries get old and damaged. Third, I will show you how we check the battery's health. Finally, I will discuss the models we use and the future of this technology.

Slide 3: Introduction

Why is State of Health, or SOH, important? It keeps Electric Vehicles safe. Most research looks at only one battery cell. But real cars use Battery Packs. A pack has many cells connected together. The problem is the "Barrel Effect." Think of a wooden barrel. It can only hold water up to its shortest piece of wood. A battery pack is the same. Its power is limited by the weakest cell. This difference between cells makes the pack age faster.

Slide 4: How Lithium-ion Batteries Work

To understand the damage, we must know how the battery works. Please look at Figure 1. It shows the main idea. Lithium ions move back and forth. They go between the positive and negative sides. When we charge, ions move to the anode. When we discharge, they move back. This movement stores the energy.

Slide 5: Cell-Level Degradation Mechanisms

This process is not perfect. Figure 2 shows the damage inside a cell. We have chemical problems, like the growth of the SEI film. This layer grows and increases resistance. We also have mechanical problems. The particles inside expand and shrink. This causes stress and breaks the structure over time.

Slide 6: Impact of Fast Charging

Fast charging makes these problems worse. High currents create heat. This heat speeds up the chemical damage. A big risk is "Lithium Plating." Lithium builds up on the surface instead of going inside. This creates "dendrites," which can cause short circuits. This is very dangerous, especially in cold weather.

Slide 7: Degradation Mechanism of Battery Packs

Now, let's look at the pack level. Please look at Figure 3 at the bottom. This simulates the "Barrel Effect" I talked about. You can see the weakest cell limits the whole pack. Figure 4 shows a "Vicious Cycle." Small differences in manufacturing cause uneven heat. This makes some cells age faster. Then, the difference becomes bigger. It is a loop that creates failure.

Slide 8: Health Feature Extraction - ICA

How do we measure health? One good way is ICA. Please look at the curves in Figure 5. The shape of the curve changes as the battery gets old. The peaks move and drop down. These peaks tell us the health status. However, this method has a problem. It is very sensitive to noise, so we need clean data.

Slide 9: Health Feature Extraction - Time & Frequency

We can also use Time and Frequency methods. For Time, we just measure the charging speed. Old batteries take longer to charge. It is simple. For Frequency, look at Figure 6. We use a method called EIS. It sees inside the chemistry clearly. But, the machine is expensive and slow.

Slide 10: Feature Comparison Table

Let's compare these methods in Table 1. We see three main points. First, ICA and Time methods are simple. They only need voltage and current data. But noise affects them easily. Second, the Frequency method gives great data, but it costs too much money and time. Third, measuring "Inconsistency" is the best way for packs. But it is very hard to check this online in real-time.

Slide 11: SOH Estimation Models - Statistical Filters

Now we have the data. We need a model to calculate health. First, we have Statistical Filters, like the Kalman Filter. This is a "White Box" method. It uses math equations. The good point is speed. It is very fast. The bad point is that it needs a perfect starting model to work well.

Slide 12: SOH Estimation Models - Data-Driven

Next is the Data-Driven approach, or "Black Box." Please look at Figure 8. We use Neural Networks here. The computer learns the health directly from data. We have a few types. SVM is good for small data. RVM and GPR are great because they tell us the confidence level. ELM is the fastest one to train. These models are very precise.

Slide 13: SOH Estimation Models - Hybrid

Finally, we have Hybrid Methods. This mixes the two types. For example, we use a Kalman Filter with a Neural Network. This gives the best accuracy. But, as you can guess, it uses a lot of computer power.

Slide 14: Comparison of Models (Qualitative)

Here is a simple comparison. Filters are the "Fast" choice. Machine Learning is the "Precise" choice. Hybrid Methods are the "Balanced" choice. Hybrid gives the best result if your computer is strong enough.

Slide 15: Table 2 Comparison

Let's look at Table 2. It shows the choice: speed versus accuracy. Filters are fast for real-time chips but less accurate. Machine Learning is very accurate but complex. This is why Hybrid Methods are the future. They balance speed and performance perfectly.

Slide 16: Estimation Error Comparison

Now, let's look at the numbers. Filter models usually have an error around 1.2% to 3%. Machine Learning models, like SVM, are better. They have errors under 0.5%. Advanced Hybrid methods are the best. Some have errors as low as 0.014%. This shows that advanced models give us better accuracy.

Slide 17 & 18: Table 3 Detailed Synopsis

Table 3 confirms these numbers. As we use better models, the error goes down. Early filters had errors up to 3%. Simple AI models reduced this to under 0.5%. Then, GPR is special because it works with partial data. Finally, Hybrid Methods win. They reach the lowest error of 0.014%. Combining models clearly gives the best result.

Slide 19: Challenges

Please look at Slide 19. We face a big challenge called the "Lab-to-Reality Gap." In the lab, our data is clean. But look at the second point. In the real world, data has noise and errors. Also, temperature changes a lot when we drive. Our models fail because they expect a stable environment.

Slide 20: Future Prospects

What is the future? We have three goals on Slide 20. First, we must adapt to new batteries, like Solid-State ones. Second, we will use better sensors, like Ultrasonic sensors, to look inside the battery. Third, and most importantly, we want to put Hybrid Models directly into the car's system. This will make the battery safer and last longer.

Slide 21: Conclusion

To conclude, we must focus on Battery Packs. The main problem is that cells are different from each other. The best solution is the Hybrid Model. It is accurate and efficient. Our final goal is to put these models into real cars. This ensures a safe and reliable future. Thank you.